

Spectral Admissibility in the Q_8 Cayley Graph: A Minimal Toy Model for Bounded-Flux Mode Selection

Working note – Cosmochrony programme

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Abstract

We establish the first analytically tractable instance of the spectral admissibility programme outlined in the Cosmochrony framework. On the undirected Cayley graph of the quaternion group Q_8 , with generators $S = \{\pm i, \pm j, \pm k\}$, the local bounded-flux constraint $|\partial_t \chi_v| \leq c_\chi$ induces a hierarchy of maximal admissible effective amplitudes across irreducible spectral sectors. In the linearised dispersion regime, the non-abelian irreducible sector admits a maximal effective amplitude larger than the nontrivial abelian sectors by a factor of $\sqrt{4/3} \approx 1.155$. Resolving explicitly the basis dependence in the degenerate ρ_4 eigenspace, we show that this advantage holds for the physically relevant effective amplitude $A_n = a_n \|\psi_n\|_\infty$, but not for the raw amplitude a_n . This constitutes a minimal model demonstration that bounded substrate capacity may structurally favour non-abelian relational modes, without invoking any additional dynamical selection principle.

1 Setting and Motivation

The Cosmochrony framework identifies the physical spectrum with the residue of two successive constraints acting on the relational operator $\mathcal{L}_\chi$:

1. **Topological discretisation:** the fibre structure of the non-injective projection Π organises modes into discrete winding sectors.
2. **Bounded-flux filtering:** the saturation bound $|\partial_t \chi| \leq c_\chi$ eliminates modes whose local evolution would exceed the substrate's relational capacity.

The goal of this note is to make step 2 explicit and calculable on the simplest non-trivial finite graph encoding a quaternionic algebraic structure, as a minimal discrete proxy for $SU(2)$ -type sectors.

2 The Q_8 Cayley Graph

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ denote the quaternion group of order 8. We define the undirected Cayley graph $\Gamma(Q_8, S)$ with generating set

$$S = \{\pm i, \pm j, \pm k\}. \tag{1}$$

Since S is closed under inversion ($s^{-1} \in S$ for all $s \in S$) and does not contain the identity, $\Gamma(Q_8, S)$ is a well-defined, connected, undirected, 6-regular graph on 8 vertices.

Remark 1. *The choice $S = \{i, j, k\}$ (without inverses) would yield a directed graph with an asymmetric adjacency operator. For a self-adjoint Laplacian, the full symmetric set $S = \{\pm i, \pm j, \pm k\}$ is required. The resulting graph is 6-regular, not 3-regular.*

2.1 Adjacency Operator and Graph Laplacian

The adjacency operator A of $\Gamma(Q_8, S)$ acts on functions $f: Q_8 \rightarrow \mathbb{C}$ by

$$(Af)(g) = \sum_{s \in S} f(gs) = \sum_{s \in \{\pm i, \pm j, \pm k\}} f(gs). \quad (2)$$

The combinatorial graph Laplacian is

$$\mathcal{L} = dI - A, \quad d = |S| = 6, \quad (3)$$

where I is the identity on $\ell^2(Q_8)$. The eigenvalues of $\mathcal{L}$ are $\lambda = 6 - \mu$, where μ ranges over the eigenvalues of A .

3 Spectral Decomposition via Representation Theory

Since Q_8 is a finite group, $\ell^2(Q_8) \cong \mathbb{C}[Q_8]$ decomposes as a Q_8 -module into irreducible representations. By the Peter–Weyl theorem for finite groups:

$$\mathbb{C}[Q_8] \cong \bigoplus_{\rho \in \widehat{Q_8}} V_\rho^{\oplus \dim \rho}, \quad (4)$$

where the sum runs over the five irreducible representations of Q_8 .

3.1 Irreducible Representations of Q_8

Q_8 has five irreducible representations:

Label	Dimension	$\rho(i)$	$\rho(j)$	$\rho(k)$
ρ_0 (trivial)	1	1	1	1
ρ_1	1	1	-1	-1
ρ_2	1	-1	1	-1
ρ_3	1	-1	-1	1
ρ_4 (quaternionic)	2	$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

The dimension check: $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8 = |Q_8|$. ✓

3.2 Eigenvalues of the Adjacency Operator

For any irreducible representation ρ , the adjacency operator acts as the scalar $\mu_\rho \cdot I_{V_\rho}$ on V_ρ . Since S is a union of conjugacy classes of Q_8 , the corresponding element of the group algebra is central, and Schur's lemma gives

$$\mu_\rho = \frac{1}{\dim \rho} \sum_{s \in S} \chi_\rho(s), \quad (5)$$

where χ_ρ is the character of ρ .

1D sectors. For the trivial representation ρ_0 : $\chi_{\rho_0}(s) = 1$ for all $s \in S$, so $\sum_{s \in S} \chi_{\rho_0}(s) = 6$ and

$$\mu_{\rho_0} = \frac{1}{1} \cdot 6, \quad \lambda_{\rho_0} = 6 - 6 = 0. \quad (6)$$

For ρ_1 : $\chi_{\rho_1}(\pm i) = 1$ and $\chi_{\rho_1}(\pm j) = \chi_{\rho_1}(\pm k) = -1$, so $\sum_{s \in S} \chi_{\rho_1}(s) = 2(1) + 4(-1) = -2$, giving $\mu_{\rho_1} = -2$ and $\lambda_{\rho_1} = 8$. By the symmetry of Q_8 under permutations of $\{i, j, k\}$, $\mu_{\rho_2} = \mu_{\rho_3} = -2$ and $\lambda_{\rho_2} = \lambda_{\rho_3} = 8$.

2D sector. For the quaternionic representation ρ_4 , we compute the character at each generator. The matrices $\rho_4(i)$, $\rho_4(-i)$, $\rho_4(j)$, $\rho_4(-j)$, $\rho_4(k)$, $\rho_4(-k)$ each have trace zero: for instance, $\text{tr}(\rho_4(i)) = \text{tr}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = 0$, and $\text{tr}(\rho_4(-i)) = 0$; similarly for j and k . Thus $\sum_{s \in S} \chi_{\rho_4}(s) = 0$ and

$$\mu_{\rho_4} = \frac{1}{2} \cdot 0 = 0, \quad \lambda_{\rho_4} = 6 - 0 = 6. \quad (7)$$

3.3 Summary of the Spectrum

Sector	λ_n	Multiplicity	Type
ρ_0	0	1	Trivial (zero mode)
ρ_1, ρ_2, ρ_3	8	3	Nontrivial abelian
ρ_4	6	4	Non-abelian (quaternionic)

Total dimension: $1 + 3 + 4 = 8$. ✓

4 Bounded-Flux Admissibility Condition

4.1 Setup

Let the substrate field $\chi: Q_8 \times \mathbb{R} \rightarrow \mathbb{R}$ evolve on the graph, subject to the local bounded-flux constraint

$$|\partial_t \chi_v(t)| \leq c_\chi \quad \forall v \in Q_8, \forall t. \quad (8)$$

Consider a single-mode excitation

$$\chi_v(t) = a_n \psi_n(v) \cos(\omega_n t), \quad (9)$$

where ψ_n is an eigenmode of $\mathcal{L}$ with eigenvalue λ_n , normalised in $\ell^2(Q_8)$, and $a_n > 0$ is the mode amplitude. The time derivative at vertex v is

$$\partial_t \chi_v(t) = -a_n \omega_n \psi_n(v) \sin(\omega_n t). \quad (10)$$

The constraint (8) requires

$$|\omega_n| \cdot a_n \cdot |\psi_n(v)| \leq c_\chi \quad \forall v \in Q_8. \quad (11)$$

The binding constraint is at the vertex where $|\psi_n(v)|$ is maximal:

$$|\omega_n| \cdot a_n \cdot \|\psi_n\|_\infty \leq c_\chi. \quad (12)$$

4.2 Effective Amplitude and Admissibility Envelope

Definition 1. The effective amplitude of a mode (ψ_n, a_n) is

$$A_n := a_n \|\psi_n\|_\infty. \quad (13)$$

A mode is dynamically admissible (in the linearised regime) if

$$|\omega_n| \cdot A_n \leq c_\chi. \quad (14)$$

In the linearised dispersion approximation, $\omega_n^2 \approx \lambda_n$, so the maximal admissible effective amplitude is

$$A_n^{\max} = \frac{c_\chi}{\sqrt{\lambda_n}}. \quad (15)$$

This defines the *admissibility envelope* in the (λ_n, A_n) plane.

Remark 2. The linearised dispersion $\omega_n^2 = \lambda_n$ is valid for $A_n \ll c_\chi / \sqrt{\lambda_n}$. The non-linear (Born-Infeld) correction modifies this relation near saturation and requires a modal reduction of the DBI action; it is deferred to a separate calculation. The envelope (15) should therefore be understood as the admissibility boundary in the weakly non-linear regime.

4.3 Explicit Eigenmodes in the ρ_4 Sector

The non-abelian sector ρ_4 has multiplicity 4 in $\ell^2(Q_8)$, so the quantity $\|\psi_n\|_\infty$ is not determined until an explicit basis of the eigenspace is chosen. A natural orthonormal basis is provided by the matrix coefficients of the quaternionic irreducible representation.

For a finite group G and an irreducible representation ρ , the standard Peter–Weyl basis of the corresponding isotypic component is given by

$$\psi_{uv}(g) = \sqrt{\frac{\dim \rho}{|G|}} \rho(g)_{uv}, \quad u, v \in \{1, \dots, \dim \rho\}. \quad (16)$$

For Q_8 and ρ_4 , this becomes

$$\psi_{uv}(g) = \frac{1}{2} \rho_4(g)_{uv}, \quad u, v \in \{1, 2\}, \quad (17)$$

since $\dim \rho_4 = 2$ and $|Q_8| = 8$.

Using the explicit matrices

$$\rho_4(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_4(-1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (18)$$

$$\rho_4(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \rho_4(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho_4(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad (19)$$

one finds that the diagonal modes ψ_{11}, ψ_{22} are supported on $\{\pm 1, \pm i\}$, while the off-diagonal modes ψ_{12}, ψ_{21} are supported on $\{\pm j, \pm k\}$. In all four cases,

$$|\psi_{uv}(g)| \in \left\{0, \frac{1}{2}\right\}, \quad (20)$$

hence

$$\|\psi_{uv}\|_\infty = \frac{1}{2}, \quad u, v \in \{1, 2\}. \quad (21)$$

For the one-dimensional nontrivial abelian sectors ρ_1, ρ_2, ρ_3 , the normalised eigenmodes are the corresponding characters scaled by $|Q_8|^{-1/2}$. Since these characters take values in $\{\pm 1\}$, one has

$$\|\psi_{\rho_k}\|_\infty = \frac{1}{\sqrt{8}}, \quad k = 1, 2, 3. \quad (22)$$

4.4 Basis-Resolved Admissibility

Proposition 1 (Basis-resolved admissibility on Q_8). *In the canonical Peter–Weyl basis of $\ell^2(Q_8)$, the maximal admissible effective amplitudes satisfy*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3, \quad (23)$$

whereas the corresponding maximal raw amplitudes satisfy

$$a_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6} \|\psi_{\rho_4}\|_\infty} = \frac{2}{\sqrt{6}} c_\chi \approx 0.816 c_\chi, \quad (24)$$

$$a_{\rho_k}^{\max} = \frac{c_\chi}{\sqrt{8} \|\psi_{\rho_k}\|_\infty} = c_\chi, \quad k = 1, 2, 3. \quad (25)$$

Thus the non-abelian sector is favoured in effective amplitude, but not in raw amplitude.

Proof. The effective-amplitude statement follows from $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ together with $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$. The raw-amplitude statement follows from $a_n^{\max} = A_n^{\max}/\|\psi_n\|_\infty$, using $\|\psi_{\rho_4}\|_\infty = 1/2$ and $\|\psi_{\rho_k}\|_\infty = 1/\sqrt{8}$. $\square$

Remark 3 (Interpretation). *The bounded-flux constraint acts on the local quantity $|\omega_n| a_n |\psi_n(v)|$, and therefore selects modes through the effective amplitude $A_n = a_n \|\psi_n\|_\infty$, not through the raw amplitude a_n alone. The non-abelian advantage on Q_8 is therefore not a larger absolute amplitude capacity, but a larger admissible effective window per unit spectral weight. Equivalently, the ρ_4 sector is less constrained by bounded-flux filtering in the physically relevant local variable.*

5 Main Result

Proposition 2 (Spectral admissibility hierarchy on Q_8). *In the undirected Cayley graph $\Gamma(Q_8, \{\pm i, \pm j, \pm k\})$, under the local bounded-flux constraint (8) and linearised dispersion, the maximal admissible effective amplitude satisfies*

$$A_{\rho_4}^{\max} = \frac{c_\chi}{\sqrt{6}} > \frac{c_\chi}{\sqrt{8}} = A_{\rho_k}^{\max}, \quad k = 1, 2, 3. \quad (26)$$

The non-abelian sector ρ_4 occupies a strictly larger admissible region in the space of effective mode amplitudes than any nontrivial abelian sector, by a factor

$$\frac{A_{\rho_4}^{\max}}{A_{\rho_k}^{\max}} = \sqrt{\frac{8}{6}} = \sqrt{\frac{4}{3}} \approx 1.155. \quad (27)$$

Proof. Direct substitution of $\lambda_{\rho_4} = 6$ and $\lambda_{\rho_k} = 8$ into the admissibility envelope (15). $\square$

Remark 4. *The trivial sector ρ_0 has $\lambda = 0$ and thus formally unbounded $A^{\max}$. It corresponds to the uniform zero mode of the connected graph. In the present analysis it is factored out, since we are interested only in non-trivial excitations above the connected background.*

6 Interpretation

Proposition 1, together with the basis-resolved Proposition ??, establishes that, at fixed substrate capacity c_χ , the non-abelian irreducible sector of the quaternion Cayley graph sustains a larger local effective amplitude before violating the bounded-flux constraint, even though its raw amplitude bound is lower than that of the nontrivial abelian sectors.

In the language of the Cosmochrony programme:

- The admissibility envelope $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ defines a *geometry of exclusion* in the space of relational modes: modes above the envelope are dynamically inadmissible.
- The non-abelian sector ρ_4 lies strictly below this envelope for a larger range of effective amplitudes than the nontrivial abelian sectors ρ_1, ρ_2, ρ_3 : the bounded-flux constraint is less restrictive for non-abelian modes, enlarging their admissible window by a factor of $\sqrt{4/3}$.
- This advantage is a statement about the physically relevant local variable $A_n = a_n \|\psi_n\|_\infty$, not about the raw amplitude a_n . In the canonical basis, the ρ_4 modes are more localised ($\|\psi\|_\infty = 1/2$) than the abelian ones ($\|\psi\|_\infty = 1/\sqrt{8}$), so the non-abelian sector is more constrained in raw amplitude even while remaining less constrained in effective amplitude.
- This provides a minimal admissibility advantage for non-abelian relational modes, without any dynamical postulate beyond the local flux bound.

Remark 5 (Real field and complex eigenmodes). *The substrate field χ is taken real-valued. The natural eigenmodes of the 2D sector are matrix coefficients of ρ_4 , which are complex-valued. One may either work in $\ell^2(Q_8, \mathbb{C})$ and take real and imaginary parts to obtain real eigenmodes, or directly choose a real basis for each eigenspace. The local admissibility criterion is intrinsically defined, but its expression in terms of (a_n, ψ_n) depends on the chosen basis within a degenerate eigenspace through $\|\psi_n\|_\infty$. This basis dependence is resolved explicitly in Section ??.*

This result does not yet establish:

- that the non-abelian sector will dominate in a full dynamical simulation;
- the origin of the eigenvalue gap $\lambda_{\rho_4} = 6 < 8 = \lambda_{\rho_k}$ from the topological structure of Π alone;
- the Born-Infeld correction to the admissibility envelope near saturation.

These remain open and constitute the next steps of the programme.

7 Next Steps

The natural continuation of this calculation proceeds as follows.

Step 2: modal DBI reduction. For a single mode $\chi_v(t) = q(t)\psi_n(v)$, substitute into the discrete DBI Lagrangian

$$L = -c_\chi^2 \sum_v \sqrt{1 - \dot{\chi}_v^2/c_\chi^2} - \frac{1}{2} \sum_{v,w} \chi_v \mathcal{L}_{vw} \chi_w, \quad (28)$$

and derive the effective equation for $q(t)$. Extract the amplitude-dependent frequency $\omega_n(a_n)$ via harmonic balance. This yields the corrected admissibility envelope replacing (15).

Step 3: dynamical simulation. Simulate the full Q_8 substrate with the DBI constraint imposed locally at each vertex. Track the modal decomposition $A_n(t) = a_n(t)\|\psi_n\|_\infty$ over time. Test whether modes initialised above the envelope (15) decay, and whether modes in the non-abelian sector survive longer than those in nontrivial abelian sectors at comparable initial A_n .

Step 4: extension to larger graphs. Replace Q_8 by the Cayley graph of a group approximating S^3 more faithfully at large N (e.g. the binary icosahedral group $2I$ of order 120, or $\text{SL}(2, \mathbb{F}_p)$ for small primes p). Test whether the spectral admissibility hierarchy persists under this refinement. A particularly natural continuation is provided by Cayley graphs in Ramanujan families – such as the Lubotzky–Phillips–Sarnak construction on $\text{SL}(2, \mathbb{F}_p)$ – where the spectral gap is asymptotically optimal. This would allow the admissibility hierarchy to be tested in graph sequences with extremal expansion properties, directly connecting bounded-flux filtering to the theory of expander graphs.